

Poisson Desingularization via Weighted Blowups

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I. Introduction

Definition. A **Poisson triple** (X, Y, π) is a complex manifold X , analytic subvariety Y , and Poisson structure π tangent to both X and Y .

Definition. A **reduction of singularities** of (X, Y, π) is a (proper, holomorphic surjective) Poisson map

$$\sigma: (X', Y', \pi') \longrightarrow (X, Y, \pi)$$

where (X', Y', π') is a Poisson triple with strictly “better” singularities and $\sigma|_{X' \setminus \sigma^{-1}(Y_{\text{sing}})}$ is an isomorphism. If Y' is smooth, this is called a **Poisson resolution**. Here “better” singularities means the singularity invariant of [1] is smaller.

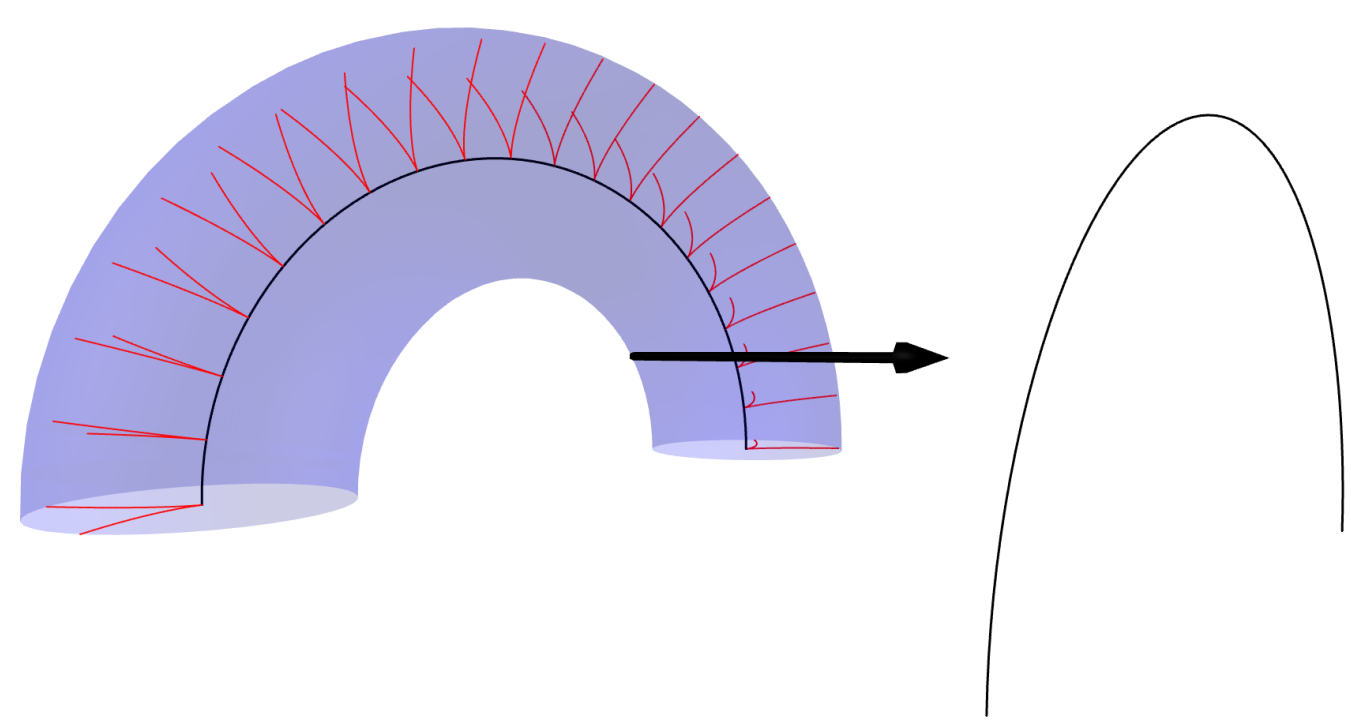
We prove the existence of reduction of Poisson singularities using weighted blowups, following the strategy of [1, 3] for classical non-Poisson desingularization (the case $\pi = 0$). In their work, weighted blowups are used to diminish a singularity invariant to some minimal value, at which point resolution is achieved. We show that for Poisson singularities this is possible up to a point. In particular, we give a complete classification of Poisson subvarieties of Poisson 3-folds which cannot be blown-up; in dimension 4 we provide a similar classification for Poisson subvarieties of Lie-Poisson (linear) structures.

II. Weighted Blowups

INPUT: a **weighted center** $Z_{\bullet} = (x_1^{1/w_1}, \dots, x_k^{1/w_k})$, which consists of

- A submanifold $Z \subset Y_{\text{sing}} \subset X$
- Submanifold coordinates $(x_1, \dots, x_n)$ s.t. $Z = \{x_1 = \dots = x_k = 0\}$
- Weights $w_1, \dots, w_n$, associated to $x_1, \dots, x_k$

OUTPUT: the **weighted blowup** $\text{Bl}_{Z_{\bullet}}(X) \rightarrow X$; an orbifold obtained by replacing Z with $E := \mathbb{P}(NZ_{\bullet})$, the “weighted projectivization” of its normal bundle.



III. Lifting Criterion

As in the non-weighted case, Poisson structures do not always lift to weighted blowups. This depends on the **weighted order**, a valuation defined by the rules:

$$\text{ord}_{Z_{\bullet}} x_i = w_i, \text{ord}_{Z_{\bullet}} \partial_{x_i} = -w_i.$$

Example.

$$\begin{aligned} \pi &= x_1 x_2^3 \partial_{x_3} \wedge \partial_{x_4} \\ \implies \text{ord}_{Z_{\bullet}} \pi &= w_1 + 3w_2 - w_3 - w_4. \end{aligned}$$

Lemma ([2]). A Poisson structure π lifts to a weighted blowup $\text{Bl}_{Z_{\bullet}}(X)$ with lift tangent to E if and only if

$$\text{ord}_{Z_{\bullet}} \pi \geq 0.$$

This is the weighted analogue of Polischuk’s criterion [4], which is retrieved by setting $w_1 = \dots = w_k = 1$.

IV. Main Theorem in 3D

Theorem ([2]). For any Poisson triple (X, Y, π) with $\dim X \leq 3$, there exists a sequence of weighted blowups

$$(X', Y', \pi') \longrightarrow \dots \longrightarrow (X, Y, \pi)$$

such that (X', Y', π') is either smooth, or has one of the following singularities, which can’t be blown-up:

► **non-nilpotent singularity:**

$$Y' = \{x = g(y, z) = 0\}, \pi' = x \partial_x \wedge \partial_y$$

► **du Val singularity:**

$$Y' = \{f = 0\}, \pi' = g \cdot [\partial_x \wedge \partial_y \wedge \partial_z, f]$$

where $g(0) = 1$ and f is one of the functions given by the ADE classification in the table below.

Name	f
$A_n, n \geq 1$	$x^2 + y^2 + z^{n+1}$
$D_n, n \geq 4$	$x^2 + y^2 z + z^{n-1}$
E_6	$x^2 + y^3 + z^4$
E_7	$x^2 + y^3 + y z^3$
E_8	$x^2 + y^3 + z^5$

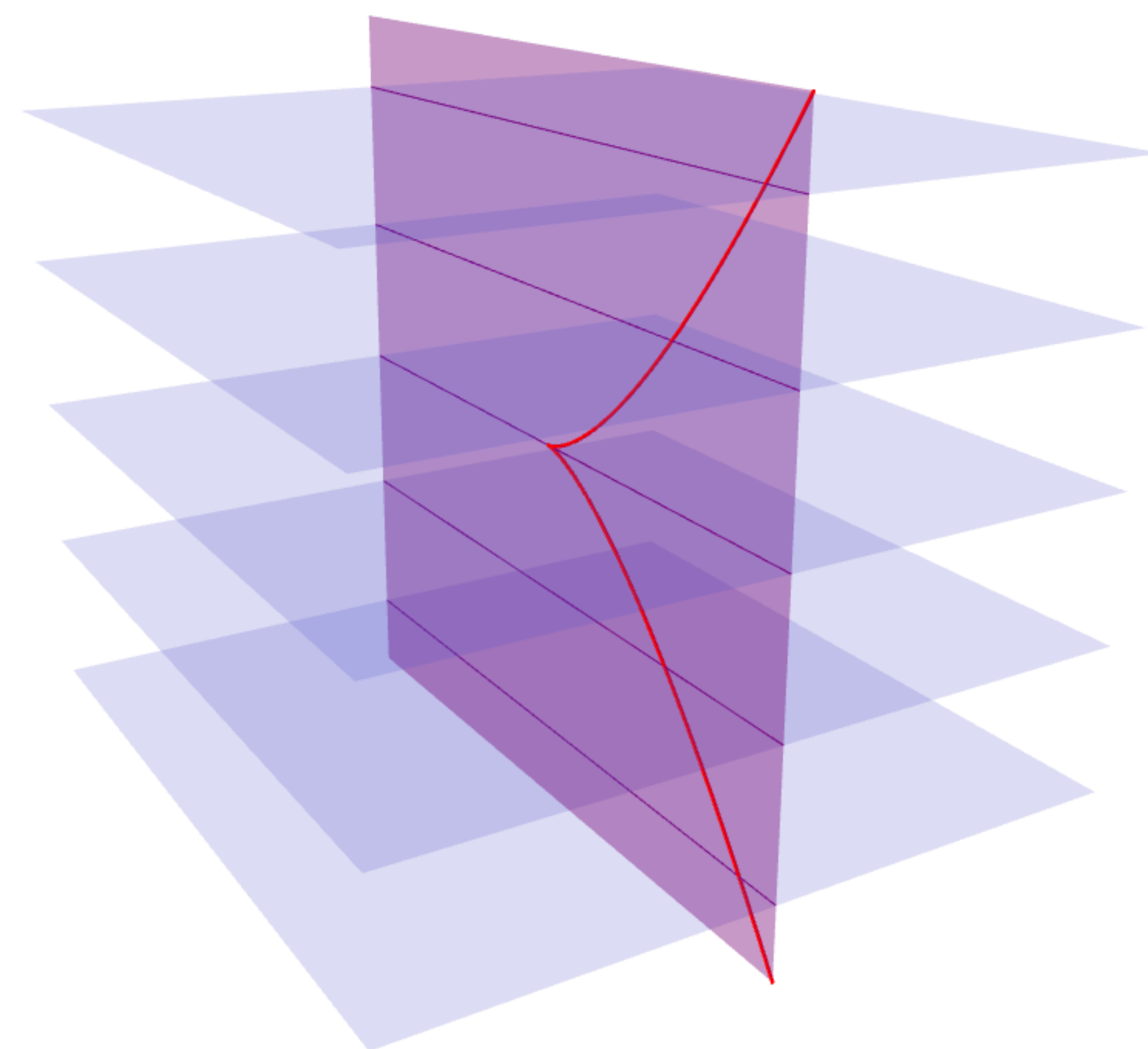


Figure: A non-nilpotent singularity (red) in the vanishing set of π (purple), and the symplectic leaves (horizontal blue).

Proof Sketch. The non-Poisson algorithm of [1] prescribes a weighted center $Z_{\bullet}$ which is guaranteed to make the singularity invariant drop after weighted blowup. If it satisfies the Poisson lifting criterion, we use it. Otherwise, we use Poisson cohomology to compute the normal form, and find a different weighted center that does. E.g. when $\dim X = 3$, $\dim Y = 2$:

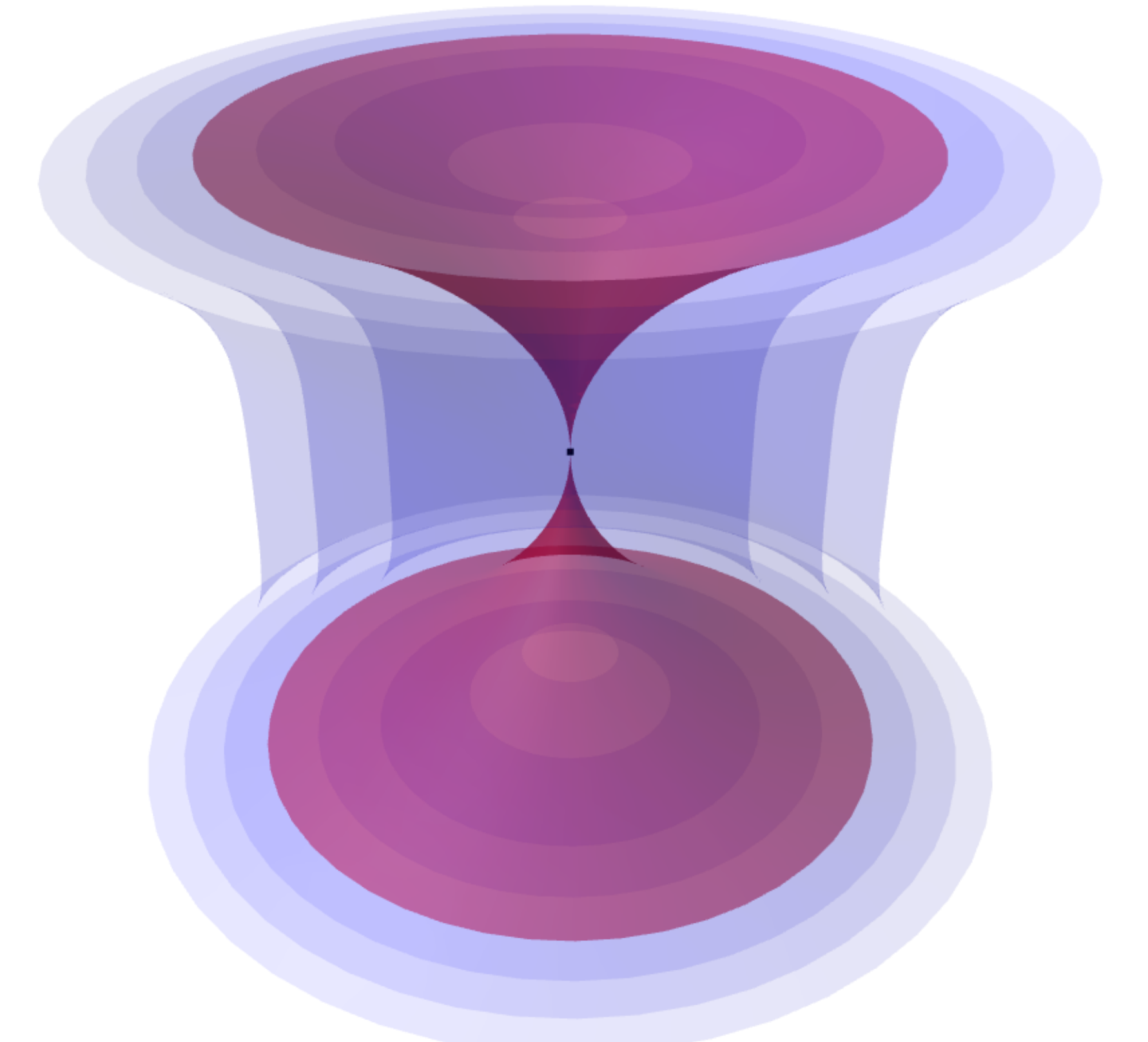
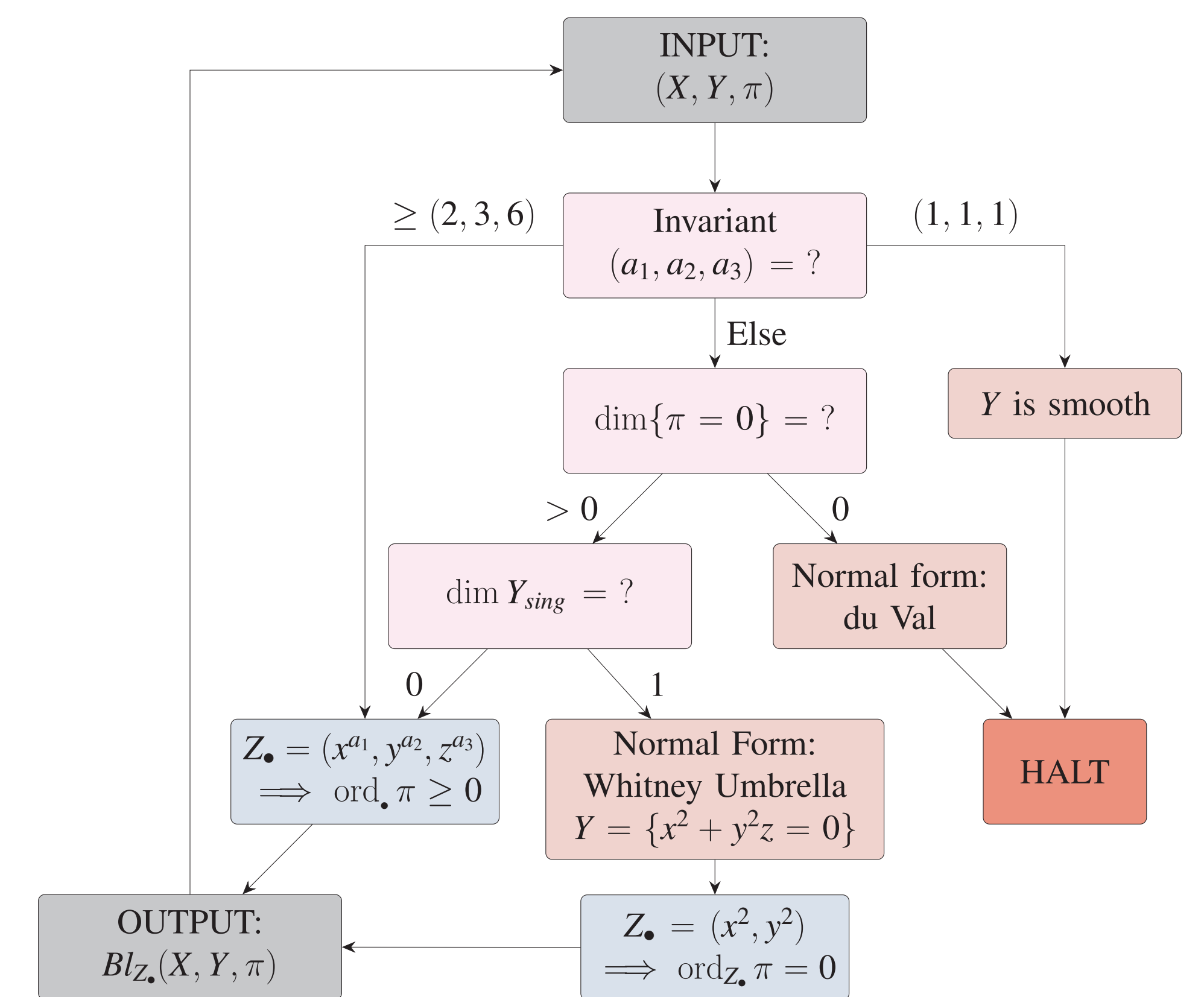


Figure: An A_3 surface singularity (dark purple), and the surrounding symplectic leaves.

V. Preliminaries in 4D: Lie-Poisson Subvarieties

Theorem (Preliminary). If $(X, \pi) = \mathfrak{g}^*$ is a 4D complex Lie-Poisson manifold, then the only singular, proper, non-trivial Poisson triples (X, Y, π) not admitting a weighted blowup fall under one of the types in the table below.

$\mathfrak{g}$	$\dim Y$	Y
$\mathfrak{aff}(1) \times \mathfrak{aff}(1)$	1	A curve in $\{\pi = 0\}$
$\mathbb{C} \ltimes_A \mathbb{C}^3$, $A \in \text{End}(\mathbb{C}^3)$	1	A curve in $\{\pi = 0\}$
	2	A surface in $\{\pi = 0\}$
$\mathfrak{gl}_2 = \mathfrak{sl}_2 \times \mathbb{C}$	2	The nilpotent cone of $\mathfrak{sl}_2$
	3	A hypersurface with the nilpotent cone of $\mathfrak{sl}_2$ as a slice
$\pi = [\partial_x \wedge \partial_y \wedge \partial_z, xw - yz]$	3	A hypersurface $\{g(xy - yz, w) = 0\}$

References

- [1] Dan Abramovich, Michael Temkin, and Jarosław Włodarczyk. Functorial embedded resolution via weighted blowings up. *Algebra & Number Theory*, 18(8):1557–1587, September 2024.
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- [3] Michael McQuillan. Very functorial, very fast, and very easy resolution of singularities. *Geometric and Functional Analysis*, 30(3):858–909, June 2020.
- [4] Alexander Polishchuk. Algebraic Geometry of Poisson Brackets. *Journal of Mathematical Sciences*, 84(5):1413–1444, 1997.

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Preprint Link

